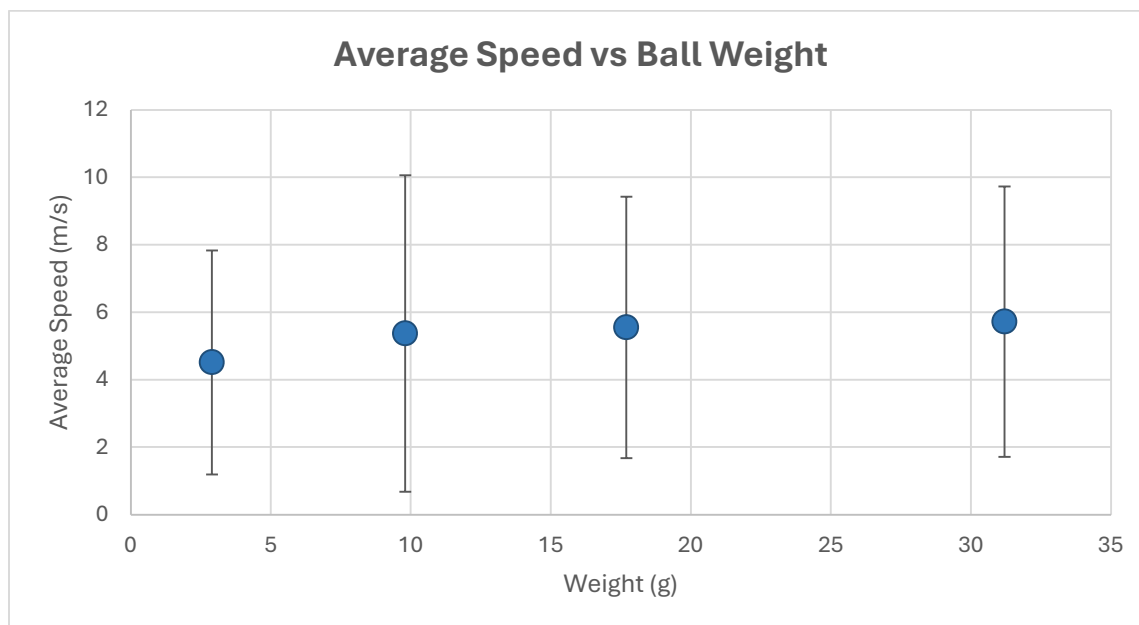


Data Quality and Experimental Limitations

Our data showed major structural problems. The trials had large run-to-run inconsistencies, and the experimental uncertainty was much larger than expected. We believe these issues came from two main sources: first, some of the balls were too heavy for the drop height used in the experiment; second, our video-based measurement method may have introduced timing errors. In particular, the iPhone 15 Pro slow-motion footage and/or the analysis software may not have produced consistent frame timing across all trials.

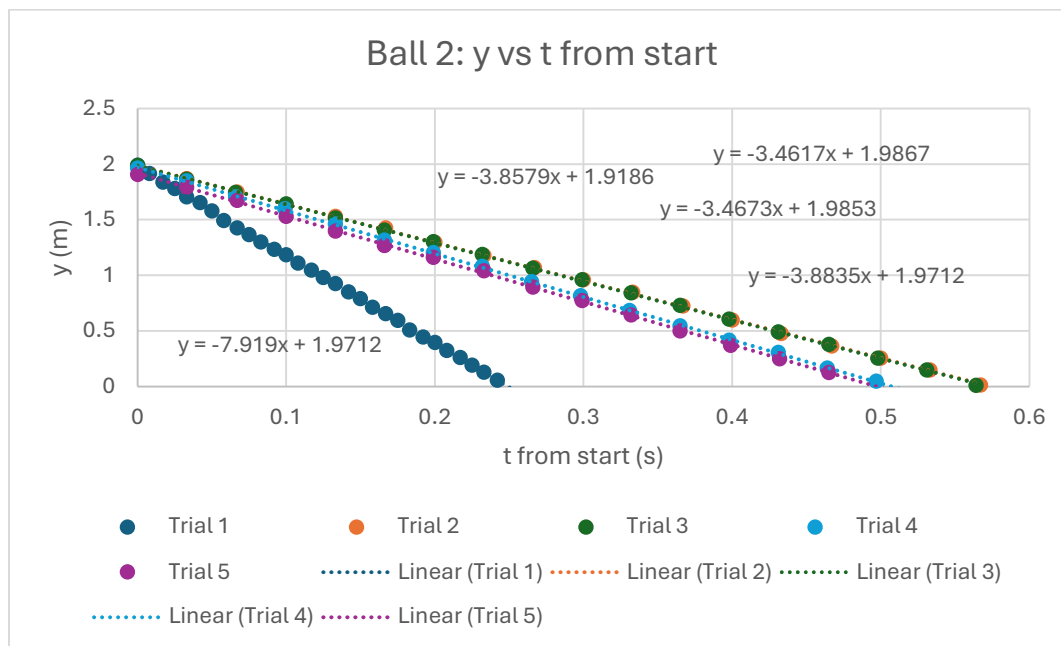
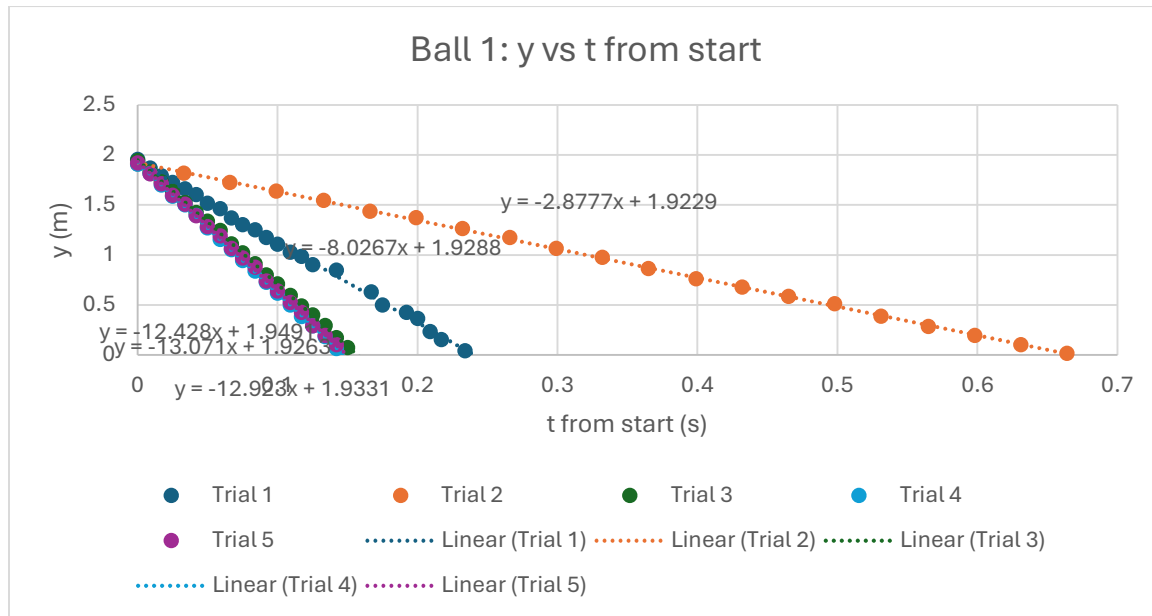
Our first approach was to plot average speed against ball mass. To calculate average speed, we measured the time it took each ball to travel through the final 2 meters before hitting the floor, and used this to compute the magnitude of the average velocity. We then averaged the speed over five trials for each ball and calculated the corresponding experimental uncertainty. However, the resulting graph was clearly unreliable: the error bars were extremely large, and the trend did not match the expected physical behavior. This led us to inspect the raw position-time data more closely.



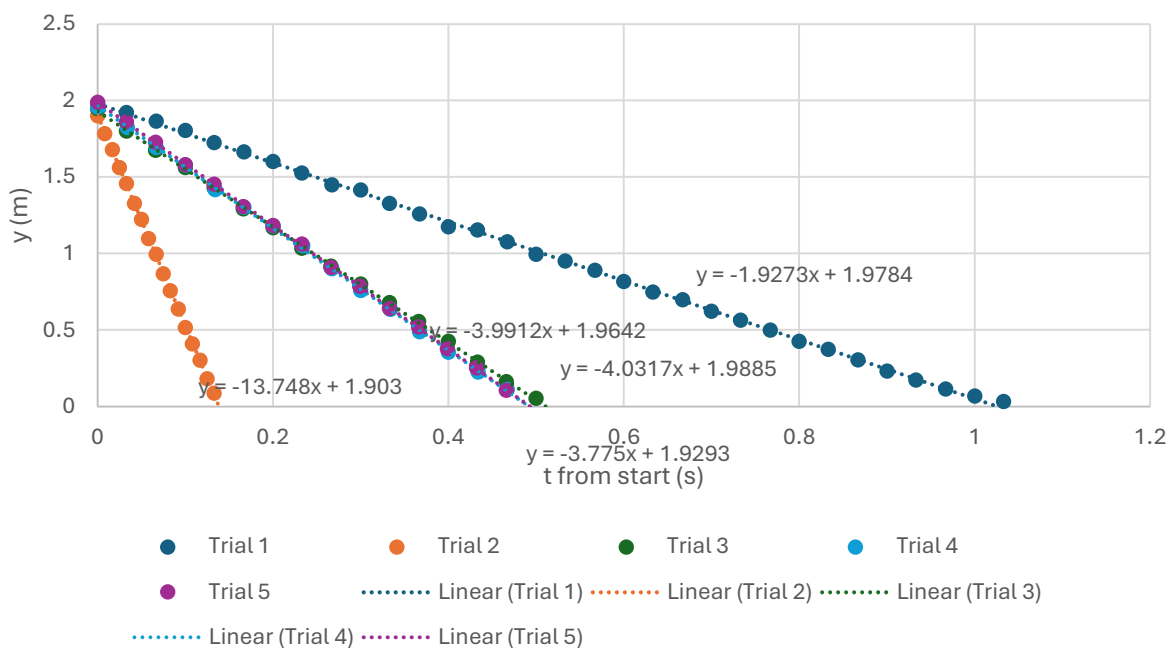
Only four balls were included in this part of the analysis because one of the balls broke during the experiment.

To investigate the source of the inconsistency, we created position-versus-time scatterplots for each ball. For each trial, we normalized the starting position to

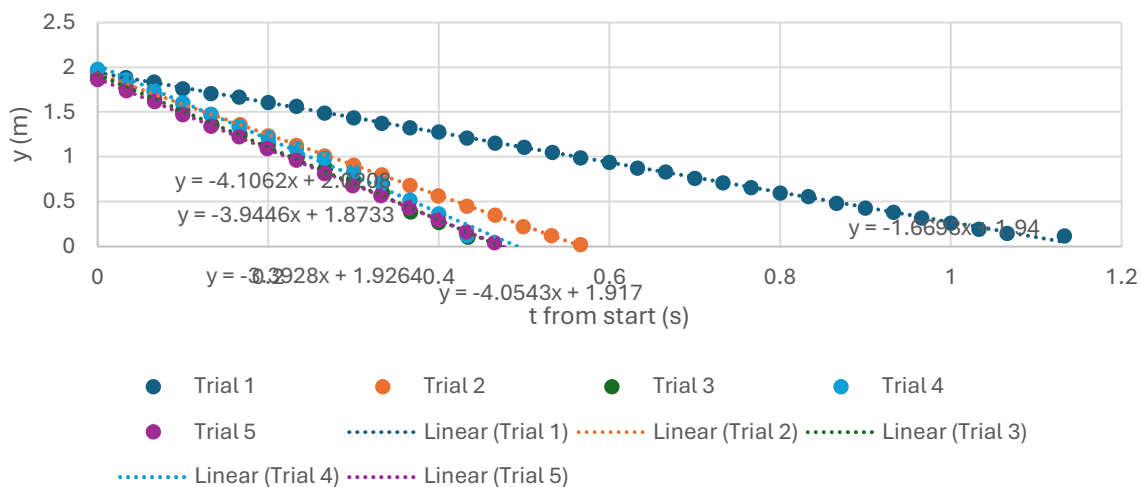
approximately 2 meters and set the initial time to $t = 0$. These plots revealed the main flaw in the data: the measured speed varied dramatically between trials, even when the same ball was dropped from the same height.

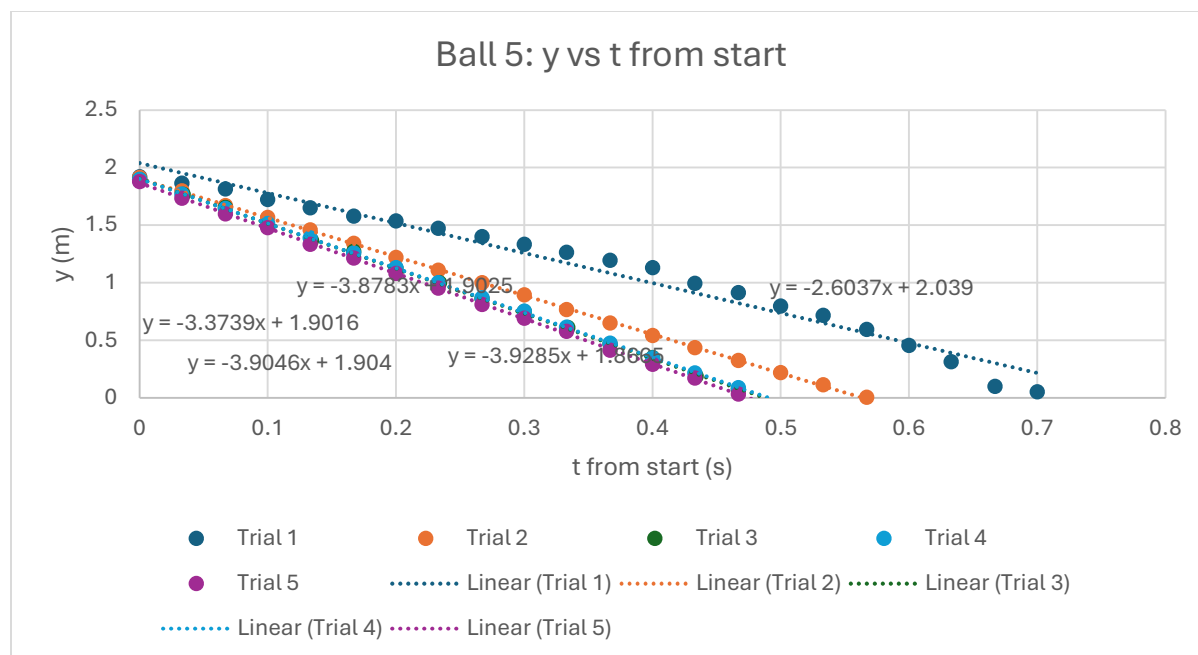


Ball 3: y vs t from start



Ball 4: y vs t from start





As seen in the graphs, the speed at which each ball moved changed substantially from trial to trial. For example, in the data for Ball 1, three trials produced slopes corresponding to speeds of about 13 m/s, while another trial gave a speed of about 8 m/s and another gave a speed of only about 2.8 m/s. Since the drop height and ball mass were held constant, variation this large is unlikely to be caused by real physical differences in the motion. Instead, it suggests that the measurement or analysis method introduced significant error.

One likely explanation is inconsistent timing in the video-analysis process. The iPhone 15 Pro officially supports slow-motion video at 1080p at either 120 fps or 240 fps, but that specification alone does not guarantee that every exported or analyzed frame is handled with perfectly uniform timing. Apple lists those recording modes, while video-production sources note that iPhone footage can involve variable-frame-rate behavior or nonstandard timing in post-production workflows. Therefore, a more cautious conclusion is that the phone/software workflow may have introduced timing errors, especially if the analysis assumed a perfectly constant frame interval. (Apple Support)

Theoretical Comparison

To better understand what speeds we should have expected, we calculated the terminal velocity of each ball using the formula below, where m is the mass of the ball, g is

gravitational acceleration, ρ is the density of air, C^d is the drag coefficient, and A is the cross-sectional area of the ball. Using $C^d = 0.47$, approximately the drag coefficient of a smooth sphere, we obtained the following terminal velocities:

$$v_t = \sqrt{\frac{2mg}{\rho C_d A}}$$

Mass	Terminal Velocity (m/s)	Terminal Velocity (km/h)
2.9 g	8.9 m/s	31.9 km/h
9.8 g	16.3 m/s	58.7 km/h
17.7 g	21.9 m/s	78.9 km/h
31.2 g	29.1 m/s	104.7 km/h

For a drop height of approximately 15 m, the expected speed after falling distance h with quadratic drag is given by the formula below. This gives:

$$v(h) = v_t \sqrt{1 - e^{-2gh/v_t^2}}$$

Mass	Expected Speed (m/s)	% of Terminal Velocity
2.9 g	8.8 m/s	98.8%
9.8 g	13.3 m/s	81.8%
17.7 g	14.8 m/s	67.7%
31.2 g	15.8 m/s	54.2%

These calculations show that the lightest ball nearly reaches terminal velocity during the drop, while the heavier balls do not. Therefore, the experiment would be more sensitive to mass differences if lower-mass balls or a larger drop height were used.

However, even after accounting for this, the theoretical values do not fully explain the large inconsistencies observed in the graphs. Since our video analysis focused on the final 2 meters of motion, we also calculated the expected speed after 13 m of falling, which is approximately where the analyzed region begins:

Mass	Expected Speed (m/s)	Speed (km/h)	% of Terminal Velocity
2.9 g	8.69 m/s	31.3 km/h	98.0%
9.8 g	12.81 m/s	46.1 km/h	78.6%
17.7 g	14.07 m/s	50.6 km/h	64.2%
31.2 g	14.84 m/s	53.4 km/h	51.0%

These speeds are still much more consistent than the speeds obtained from our experimental position-time graphs. This suggests that the main problem was not the physical model itself, but rather the quality of the experimental measurements. In

particular, the very large run-to-run differences point toward timing errors, frame-rate issues, calibration errors, or software-analysis problems.